Subject : Burgers equation
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1 Burgers equation

The Burgers equation reads (Debnath (2008)):

$$\boxed{\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} - \nu\frac{\partial^2 u}{\partial x^2} = 0} \tag{1}$$

Because we will use a finite volume method to solve this equation, it is used in the conservative form, i.e.:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} - \frac{\partial}{\partial x}\left(\nu\frac{\partial u}{\partial x}\right) = 0 \tag{2}$$

Applying the finite volume approach:

$$\int_\Omega \frac{\partial u}{\partial t}\,d\Omega + \frac{1}{2}\int_\Omega \frac{\partial u^2}{\partial x}\,d\Omega \; - \int_\Omega \frac{\partial}{\partial x}\left(\nu\frac{\partial u}{\partial x}\right)\,d\Omega = 0 \tag{3}$$

and applying Green's theorem yields:

$$\int_\Omega \frac{\partial u}{\partial t}\,d\Omega + \frac{1}{2}\oint_\Gamma u^2\,d\Gamma - \oint_\Gamma \nu\frac{\partial u}{\partial x}\,d\Gamma = 0 \tag{4}$$

2 Discretization in time and space

The discretization in space and the discretization in space are discussed in separate sections. Due to the applying regularization procedure the viscosity (ν) is increased by an artificial viscosity (Ψ), so the the implementation of $\nu + \Psi$ is a function of x.

2.1 Discretization in time

The used discretization in time reads:

$$\begin{aligned}\int_\Omega \frac{\partial u}{\partial t}\,d\Omega \Rightarrow \Delta x\Delta t_{inv}\left(\frac{1}{8}\Delta u_{i-1}^{n+1,p+1} + \frac{6}{8}\Delta u_{i}^{n+1,p+1} + \frac{1}{8}\Delta u_{i+1}^{n+1,p+1}\right) = \\ = -\Delta x\Delta t_{inv}\left(\frac{1}{8}\left(u_{i-1}^{n+1,p} - u_{i-1}^{n}\right) + \frac{6}{8}\left(u_{i}^{n+1,p} - u_{i}^{n}\right) + \frac{1}{8}\left(u_{i+1}^{n+1,p} - u_{i+1}^{n}\right)\right)\end{aligned} \tag{5}$$

2.2 Discretization in place

Convection-term

$$\frac{1}{2}\oint_\Gamma u^2 \, d\Gamma = \frac{1}{2}\left. u^2\right|_{i+\frac{1}{2}} - \frac{1}{2}\left. u^2\right|_{i-\frac{1}{2}} \tag{6}$$

with the approximation $u^2 = (u^{n+\theta,p})^2 + 2u^{n+\theta,p}\theta\Delta u$ the discretization reads:

$$\approx u_{i+\frac{1}{2}}^{n+\theta,p}\theta\Delta u_{i+\frac{1}{2}} - u_{i-\frac{1}{2}}^{n+\theta,p}\theta\Delta u_{i-\frac{1}{2}} + \frac{1}{2}\left(u_{i+\frac{1}{2}}^{n+\theta,p}\right)^2 - \frac{1}{2}\left(u_{i-\frac{1}{2}}^{n+\theta,p}\right)^2 \tag{7}$$

Viscosity term

After linearization of $c^{n+\theta,p+1}$ the discretization and with $\varepsilon = \nu + \Psi$ reads:

$$-\oint_\Gamma \varepsilon\frac{\partial u}{\partial x}\, d\Gamma = \left(\varepsilon\frac{\partial u}{\partial x}\right)_{i+\frac{1}{2}} - \left(\varepsilon\frac{\partial u}{\partial x}\right)_{i-\frac{1}{2}} \tag{8}$$

$$\Rightarrow \theta\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x}\Delta u_{i+1}^{n+1,p+1} - \theta\left(\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x} + \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x}\right)\Delta u_i^{n+1,p+1} - \theta\frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x}\Delta u_{i-1}^{n+1,p+1} + \tag{9}$$

$$+\left(\varepsilon_{i+\frac{1}{2}}\frac{u_{i+1}^{n+\theta,p} - u_i^{n+\theta,p}}{\Delta x} - \varepsilon_{i-\frac{1}{2}}\frac{u_i^{n+\theta,p} - c_{i-1}^{n+\theta,p}}{\Delta x}\right)\Bigg\} \tag{10}$$

References

Debnath, Lokenath (2008). *Sir Lighthill and Modern Fluid Dynamics*. Imperial College Press.